Numerical Sound Synthesis

Michael Dzjaparidze

December 2023

1 Preliminaries

1.1 Shift, difference and averaging operators

The forward and backward shifts and the identity operation "1" are defined as

$$e_{t+}u_d^n = u_d^{n+1} \qquad e_{t-}u_d^n = u_d^{n-1} \qquad 1u_d^n = u_d^n$$

The first-derivative operator may be approximated as

$$\delta_{t+} \triangleq \frac{1}{k}\left(e_{t+} - 1\right) \approxeq \frac{d}{dt} \tag{1a}$$

$$\delta_{t-} \triangleq \frac{1}{k}\left(1 - e_{t-}\right) \approxeq \frac{d}{dt} \tag{1b}$$

$$\delta_{t\cdot} \triangleq \frac{1}{2k}\left(e_{t+} - e_{t-}\right) \approxeq \frac{d}{dt} \tag{1c}$$

Also useful, especially in the construction of so-called implicit schemes and in energetic analysis are various averaging operators:

$$\mu_{t+} \triangleq \frac{1}{2}\left(e_{t+} + 1\right) \approxeq 1 \tag{2a}$$

$$\mu_{t-} \triangleq \frac{1}{2}\left(1 + e_{t-}\right) \approxeq 1 \tag{2b}$$

$$\mu_{t\cdot} \triangleq \frac{1}{2}\left(e_{t+} + e_{t-}\right) \approxeq 1 \tag{2c}$$

1.2 (Product) identities and inequalities

Listed below are various (product) identities which are of use in the energetic analysis of finite difference schemes.

$$\mu_{t\cdot} = 1 + \frac{k^2}{2}\delta_{tt} \tag{3a}$$

$$\delta_{t\cdot} = \delta_{t+}\mu_{t-} = \delta_{t-}\mu_{t+} \tag{3b}$$

$$\delta_{tt} = \frac{1}{k}\left(\delta_{t+} - \delta_{t-}\right) \tag{3c}$$

$$1 = \mu_{t\pm} \mp \frac{k}{2}\delta_{t\pm} \tag{3d}$$

$$e_{t\pm} = \mu_{t\pm} \pm \frac{k}{2}\delta_{t\pm} \tag{3e}$$

$$e_{t\pm} = 1 \pm k\delta_{t\pm} \tag{3f}$$

$$\delta_{tt} = \frac{2}{k}\left(\delta_{t\cdot} - \delta_{t-}\right) \tag{3g}$$

$$\mu_{t\cdot} = k\delta_{t\cdot} + e_{t-} \tag{3h}$$

$$\langle \delta_{t\cdot} u, \delta_{tt} u\rangle_{\mathcal{D}} = \delta_{t+}\left(\frac{1}{2}\|\delta_{t-}u\|^2_{\mathcal{D}}\right) \tag{4a}$$

$$\langle \delta_{t\cdot} u, u\rangle_{\mathcal{D}} = \delta_{t+}\left(\frac{1}{2}\langle u, e_{t-}u\rangle_{\mathcal{D}}\right) \tag{4b}$$

$$\langle u, e_{t-}u\rangle_{\mathcal{D}} = \|\mu_{t-}u\|^2_{\mathcal{D}} - \frac{k^2}{4}\|\delta_{t-}u\|^2_{\mathcal{D}} \tag{4c}$$

1.2.1 The discrete Laplacian and biharmonic operators

A discrete five point approximation to the Laplacian operator, making use of points adjacent to the center point, is of the form

$$\delta_{\Delta\boxplus} = \delta_{xx} + \delta_{yy} \tag{5}$$

Also important, in the case of the vibrating stiff plate, is the discrete biharmonic operator, or bi-Laplacian, which consists of the composition of the Laplacian with itself A simple approximation may be given as

$$\delta_{\Delta\boxplus,\Delta\boxplus} \triangleq \delta_{\Delta\boxplus}\delta_{\Delta\boxplus} = \delta_{xx}\delta_{xx} + \delta_{yy}\delta_{yy} + \delta_{xx}\delta_{yy} + \delta_{yy}\delta_{xx} \tag{6}$$

1.3 Inner products and identities

An l_2 spatial inner product of two 1D grid functions, f^n_l and g^n_l, over the interval $l \in \mathcal{D}$, may be defined as

$$\langle f^n, g^n\rangle_{\mathcal{D}} = \sum_{l\in\mathcal{D}} h f^n_l g^n_l = \sum_{l=d_-}^{d_+} h f^n_l g^n_l \tag{7}$$

From this, an l_2 norm follows as

$$\|f^n\|_{\mathcal{D}} = \sqrt{\langle f^n, f^n\rangle_{\mathcal{D}}} \geq 0 \tag{8}$$

The discrete inner product and norm over a Cartesian grid in 2D are a direct extension of those given by (7) and (8). For an arbitrary domain $\mathcal{D}$, the simplest definition, for two grid function $f_{l,m}$ and $g_{l,m}$ defined over a grid of spacing h_x in the x and h_y in the y direction, is

$$\langle f^n, g^n\rangle_{\mathcal{D}} = \sum_{(l,m)\in\mathcal{D}} h_x h_y f^n_{l,m} g^n_{l,m} = \sum_{l=x_-}^{x_+}\sum_{m=y_-}^{y_+} h_x h_y f^n_{l,m} g^n_{l,m} \qquad \|f^n\|_{\mathcal{D}} = \langle f^n, f^n\rangle_{\mathcal{D}} \geq 0 \tag{9}$$

The Cauchy-Schwartz and triangle inequalities extend directly to such discrete inner products as

$$|\langle f^n, g^n\rangle_{\mathcal{D}}| \leq \|f^n\|_{\mathcal{D}}\|g^n\|_{\mathcal{D}} \tag{10a}$$
$$\|f^n + g^n\|_{\mathcal{D}} \leq \|f^n\|_{\mathcal{D}} + \|g^n\|_{\mathcal{D}} \tag{10b}$$

Other types of inner products may be defined:

$$\langle f^n, g^n\rangle'_{\mathbb{U}_N} = \sum_{l=1}^{N-1} h f^n_l g^n_l + \frac{h}{2} f^n_0 g^n_0 + \frac{h}{2} f_N g^n_N \tag{11a}$$
$$\langle f^n, g^n\rangle_{\underline{\mathbb{U}}_N} = \sum_{l=0}^{N-1} h f^n_l g^n_l \tag{11b}$$
$$\langle f^n, g^n\rangle_{\overline{\mathbb{U}}_N} = \sum_{l=1}^{N} h f^n_l g^n_l \tag{11c}$$
$$\langle f^n, g^n\rangle_{\overline{\underline{\mathbb{U}}}_N} = \sum_{l=1}^{N-1} h f^n_l g^n_l \tag{11d}$$

which extend naturally to 2D:

$$\begin{aligned}\langle f^n, g^n\rangle'_{\mathbb{U}^2_{N_x,N_y}} = &\sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y f^n_{l,m} g^n_{l,m} + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} f^n_{l,0} g^n_{l,0} + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} f^n_{0,m} g^n_{0,m} \\ &+ \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} f^n_{l,N_y} g^n_{l,N_y} + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} f^n_{N_x,m} g^n_{N_x,m} + \frac{h_x h_y}{4} f^n_{0,0} g^n_{0,0} \\ &+ \frac{h_x h_y}{4} f^n_{0,N_y} g^n_{0,N_y} + \frac{h_x h_y}{4} f^n_{N_x,0} g^n_{N_x,0} + \frac{h_x h_y}{4} f^n_{N_x,N_y} g^n_{N_x,N_y}\end{aligned} \tag{12a}$$
$$\langle f^n, g^n\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} = \sum_{l=0}^{N_x-1}\sum_{m=0}^{N_y-1} h_x h_y f^n_{l,m} g^n_{l,m} \tag{12b}$$
$$\langle f^n, g^n\rangle_{\overline{\mathbb{U}}^2_{N_x,N_y}} = \sum_{l=1}^{N_x}\sum_{m=1}^{N_y} h_x h_y f^n_{l,m} g^n_{l,m} \tag{12c}$$
$$\langle f^n, g^n\rangle_{\overline{\underline{\mathbb{U}}}^2_{N_x,N_y}} = \sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y f^n_{l,m} g^n_{l,m} \tag{12d}$$
$$\{f^n, g^n\}_{(0,\mathbb{U}_{N_y})} = \sum_{m=0}^{N_y} h_y f^n_{0,m} g^n_{0,m} \tag{12e}$$
$$\{f^n, g^n\}'_{(0,\mathbb{U}_{N_y})} = \sum_{m=1}^{N_y-1} h_y f^n_{0,m} g^n_{0,m} + \frac{h_y}{2} f^n_{0,0} g^n_{0,0} + \frac{h_y}{2} f^n_{0,N_y} g^n_{0,N_y} \tag{12f}$$

1.4 Summation by parts

The following series manipulations are of use in the energetic analysis of finite difference schemes.

$$\langle f, \delta_{x+} g\rangle_{\underline{\mathbb{U}}_N} = -\langle \delta_{x-} f, g\rangle_{\overline{\mathbb{U}}_N} - f_0 g_0 + f_N g_N \tag{13a}$$
$$\langle f, \delta_{x-} g\rangle_{\overline{\mathbb{U}}_N} = -\langle \delta_{x+} f, g\rangle_{\underline{\mathbb{U}}_N} - f_0 g_0 + f_N g_N \tag{13b}$$
$$\langle f, \delta_{xx} g\rangle_{\mathbb{U}_N} = -\langle \delta_{x-} f, \delta_{x-} g\rangle_{\overline{\mathbb{U}}_N} - f_0 \delta_{x-} g_0 + f_N \delta_{x+} g_N \tag{13c}$$
$$\langle f, \delta_{xx} g\rangle_{\mathbb{U}_N} = -\langle \delta_{x+} f, \delta_{x+} g\rangle_{\underline{\mathbb{U}}_N} - f_0 \delta_{x-} g_0 + f_N \delta_{x+} g_N \tag{13d}$$
$$\langle f, \delta_{xx} g\rangle'_{\mathbb{U}_N} = -\langle \delta_{x-} f, \delta_{x-} g\rangle_{\overline{\mathbb{U}}_N} - f_0 \delta_{x\cdot} g_0 + f_N \delta_{x\cdot} g_N \tag{13e}$$
$$\langle f, \delta_{xx} g\rangle'_{\mathbb{U}_N} = -\langle \delta_{x+} f, \delta_{x+} g\rangle_{\underline{\mathbb{U}}_N} - f_0 \delta_{x\cdot} g_0 + f_N \delta_{x\cdot} g_N \tag{13f}$$
$$\langle f, \delta_{xx} g\rangle_{\mathbb{U}_N} = \langle \delta_{xx} f, g\rangle_{\mathbb{U}_N} - f_0 \delta_{x-} g_0 + f_N \delta_{x+} g_N + g_0 \delta_{x-} f_0 - g_N \delta_{x+} f_N \tag{13g}$$
$$\langle f, \delta_{xx} g\rangle_{\mathbb{U}_N} = \langle \delta_{xx} f, g\rangle_{\overline{\mathbb{U}}_N} - f_0 \delta_{x-} g_0 + f_N \delta_{x+} g_N + g_0 \delta_{x+} f_0 - g_N \delta_{x-} f_N \tag{13h}$$
$$\langle f, \delta_{xx} g\rangle'_{\mathbb{U}_N} = \langle \delta_{xx} f, g\rangle_{\overline{\underline{\mathbb{U}}}_N} - f_0 \delta_{x\cdot} g_0 + f_N \delta_{x\cdot} g_N + g_0 \delta_{x\cdot} f_0 - g_N \delta_{x\cdot} f_N \tag{13i}$$

$$\begin{aligned}\langle f, \delta_{\Delta\boxplus} g\rangle_{\mathbb{U}^2_{N_x,N_y}} = &-\langle \delta_{x+} f, \delta_{x+} g\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \langle \delta_{y+} f, \delta_{y+} g\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} \\ &- \{f, \delta_{y-} g\}_{(\underline{\mathbb{U}}_{N_x},0)} - \{f, \delta_{x-} g\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} + \{f, \delta_{y+} g\}_{(\underline{\mathbb{U}}_{N_x},N_y)} + \{f, \delta_{x+} g\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\end{aligned} \tag{14a}$$

$$\begin{aligned}\langle f, \delta_{\Delta\boxplus} g\rangle'_{\mathbb{U}^2_{N_x,N_y}} = &-\langle \delta_{x+} f, \delta_{x+} g\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \langle \delta_{y+} f, \delta_{y+} g\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} + \frac{h_y}{2}\{\delta_{x+} f, \delta_{x+} g\}_{(\underline{\mathbb{U}}_{N_x},0)} \\ &+ \frac{h_x}{2}\{\delta_{y+} f, \delta_{y+} g\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} - \frac{h_y}{2}\{\delta_{x+} f, \delta_{x+} g\}_{(\underline{\mathbb{U}}_{N_x},N_y)} - \frac{h_x}{2}\{\delta_{y+} f, \delta_{y+} g\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)} \\ &- \{f, \delta_{y\cdot} g\}'_{(\mathbb{U}_{N_x},0)} - \{f, \delta_{x\cdot} g\}'_{\left(0,\mathbb{U}_{N_y}\right)} + \{f, \delta_{y\cdot} g\}'_{(\mathbb{U}_{N_x},N_y)} + \{f, \delta_{x\cdot} g\}'_{\left(N_x,\mathbb{U}_{N_y}\right)}\end{aligned} \tag{14b}$$

$$\begin{aligned}\langle f, \delta_{\Delta\boxplus} g\rangle_{\mathbb{U}^2_{N_x,N_y}} = &\langle \delta_{\Delta\boxplus} f, g\rangle_{\overline{\mathbb{U}}^2_{N_x,N_y}} \\ &- \{f, \delta_{y-} g\}_{(\overline{\mathbb{U}}_{N_x},0)} + \{\delta_{y+} f, g\}_{(\overline{\mathbb{U}}_{N_x},0)} - \{f, \delta_{x-} g\}_{\left(0,\overline{\mathbb{U}}_{N_y}\right)} + \{\delta_{x+} f, g\}_{\left(0,\overline{\mathbb{U}}_{N_y}\right)} \\ &+ \{f, \delta_{y+} g\}_{(\overline{\mathbb{U}}_{N_x},N_y)} - \{\delta_{y-} f, g\}_{(\overline{\mathbb{U}}_{N_x},N_y)} + \{f, \delta_{x+} g\}_{\left(N_x,\overline{\mathbb{U}}_{N_y}\right)} - \{\delta_{x-} f, g\}_{\left(N_x,\overline{\mathbb{U}}_{N_y}\right)}\end{aligned} \tag{15a}$$

$$\begin{aligned}\langle f, \delta_{\Delta\boxplus} g\rangle_{\mathbb{U}^2_{N_x,N_y}} = &\langle \delta_{\Delta\boxplus} f, g\rangle_{\mathbb{U}^2_{N_x,N_y}} \\ &- \{f, \delta_{y-} g\}_{(\mathbb{U}_{N_x},0)} + \{\delta_{y-} f, g\}_{(\mathbb{U}_{N_x,0})} - \{f, \delta_{x-} g\}_{(0,\mathbb{U}_{N_y})} + \{\delta_{x-} f, g\}_{(0,\mathbb{U}_{N_y})} \\ &+ \{f, \delta_{y+} g\}_{(\mathbb{U}_{N_x},N_y)} - \{\delta_{y+} f, g\}_{(\mathbb{U}_{N_x},N_y)} + \{f, \delta_{x+} g\}_{(N_x,\mathbb{U}_{N_y})} - \{\delta_{x+} f, g\}_{(N_x,\mathbb{U}_{N_y})}\end{aligned} \tag{15b}$$

$$\begin{aligned}\langle f, \delta_{\Delta\boxplus} g\rangle'_{\mathbb{U}^2_{N_x,N_y}} = &\langle \delta_{\Delta\boxplus} f, g\rangle'_{\mathbb{U}^2_{N_x,N_y}} \\ &- \{f, \delta_{y\cdot} g\}'_{(\mathbb{U}_{N_x},0)} + \{\delta_{y\cdot} f, g\}'_{(\mathbb{U}_{N_x},0)} - \{f, \delta_{x\cdot} g\}'_{\left(0,\mathbb{U}_{N_y}\right)} + \{\delta_{x\cdot} f, g\}'_{\left(0,\mathbb{U}_{N_y}\right)} \\ &+ \{f, \delta_{y\cdot} g\}'_{(\mathbb{U}_{N_x},N_y)} - \{\delta_{y\cdot} f, g\}'_{(\mathbb{U}_{N_x},N_y)} + \{f, \delta_{x\cdot} g\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} - \{\delta_{x\cdot} f, g\}'_{\left(N_x,\mathbb{U}_{N_y}\right)}\end{aligned} \tag{15c}$$

$$
\begin{aligned}
\langle \delta_{xx} f, \delta_{yy} g \rangle_{\mathbb{U}^2_{N_x,N_y}} &= \langle \delta_{x+}\delta_{y+} f, \delta_{x+}\delta_{y+} g \rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \{f, \delta_{xx}\delta_{y-} g\}_{(\mathbb{U}_{N_x},0)} - \{\delta_{x-} f, \delta_{yy} g\}_{\left(0,\mathbb{U}_{N_y}\right)} \\
&\quad + \{f, \delta_{xx}\delta_{y+} g\}_{(\mathbb{U}_{N_x},N_y)} + \{\delta_{x+} f, \delta_{yy} g\}_{\left(N_x,\mathbb{U}_{N_y}\right)} \\
&\quad - f_{0,0}\delta_{x-}\delta_{y-} g_{0,0} + f_{0,N_y}\delta_{x-}\delta_{y+} g_{0,N_y} + f_{N_x,0}\delta_{x+}\delta_{y-} g_{N_x,0} - f_{N_x,N_y}\delta_{x+}\delta_{y+} g_{N_x,N_y}
\end{aligned} \tag{16a}
$$

$$
\begin{aligned}
\langle \delta_{yy} f, \delta_{xx} g \rangle_{\mathbb{U}^2_{N_x,N_y}} &= \langle \delta_{x+}\delta_{y+} f, \delta_{x+}\delta_{y+} g \rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \{\delta_{y-} f, \delta_{xx} g\}_{(\mathbb{U}_{N_x},0)} - \{f, \delta_{x-}\delta_{yy} g\}_{\left(0,\mathbb{U}_{N_y}\right)} \\
&\quad + \{\delta_{y+} f, \delta_{xx} g\}_{(\mathbb{U}_{N_x},N_y)} + \{f, \delta_{x+}\delta_{yy} g\}_{\left(N_x,\mathbb{U}_{N_y}\right)} \\
&\quad - f_{0,0}\delta_{x-}\delta_{y-} g_{0,0} + f_{0,N_y}\delta_{x-}\delta_{y+} g_{0,N_y} + f_{N_x,0}\delta_{x+}\delta_{y-} g_{N_x,0} - f_{N_x,N_y}\delta_{x+}\delta_{y+} g_{N_x,N_y}
\end{aligned} \tag{16b}
$$

$$
\begin{aligned}
\langle \delta_{xx} f, \delta_{yy} g \rangle_{\overline{\mathbb{U}}^2_{N_x,N_y}} &= \langle \delta_{x+}\delta_{y+} f, \delta_{x+}\delta_{y+} g \rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \{f, \delta_{xx}\delta_{y+} g\}_{\left(\overline{\mathbb{U}}_{N_x},0\right)} - \{\delta_{x+} f, \delta_{yy} g\}_{\left(0,\overline{\mathbb{U}}_{N_y}\right)} \\
&\quad + \{f, \delta_{xx}\delta_{y-} g\}_{\left(\overline{\mathbb{U}}_{N_x},N_y\right)} + \{\delta_{x-} f, \delta_{yy} g\}_{\left(N_x,\overline{\mathbb{U}}_{N_y}\right)} \\
&\quad - f_{0,0}\delta_{x+}\delta_{y+} g_{0,0} + f_{0,N_y}\delta_{x+}\delta_{y-} g_{0,N_y} + f_{N_x,0}\delta_{x-}\delta_{y+} g_{N_x,0} - f_{N_x,N_y}\delta_{x-}\delta_{y-} g_{N_x,N_y}
\end{aligned} \tag{16c}
$$

$$
\begin{aligned}
\langle \delta_{yy} f, \delta_{xx} g \rangle_{\overline{\mathbb{U}}^2_{N_x,N_y}} &= \langle \delta_{x+}\delta_{y+} f, \delta_{x+}\delta_{y+} g \rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \{\delta_{y+} f, \delta_{xx} g\}_{\left(\overline{\mathbb{U}}_{N_x},0\right)} - \{f, \delta_{x+}\delta_{yy} g\}_{\left(0,\overline{\mathbb{U}}_{N_y}\right)} \\
&\quad + \{\delta_{y-} f, \delta_{xx} g\}_{\left(\overline{\mathbb{U}}_{N_x},N_y\right)} + \{f, \delta_{x-}\delta_{yy} g\}_{\left(N_x,\overline{\mathbb{U}}_{N_y}\right)} \\
&\quad - f_{0,0}\delta_{x+}\delta_{y+} g_{0,0} + f_{0,N_y}\delta_{x+}\delta_{y-} g_{0,N_y} + f_{N_x,0}\delta_{x-}\delta_{y+} g_{N_x,0} - f_{N_x,N_y}\delta_{x-}\delta_{y-} g_{N_x,N_y}
\end{aligned} \tag{16d}
$$

$$
\begin{aligned}
\langle \delta_{xx} f, \delta_{yy} g \rangle'_{\mathbb{U}^2_{N_x,N_y}} &= \langle \delta_{x+}\delta_{y+} f, \delta_{x+}\delta_{y+} g \rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \{f, \delta_{xx}\delta_{y\cdot} g\}'_{(\mathbb{U}_{N_x},0)} - \{\delta_{x\cdot} f, \delta_{yy} g\}'_{\left(0,\mathbb{U}_{N_y}\right)} \\
&\quad + \{f, \delta_{xx}\delta_{y\cdot} g\}'_{(\mathbb{U}_{N_x},N_y)} + \{\delta_{x\cdot} f, \delta_{yy} g\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} \\
&\quad - f_{0,0}\delta_{x\cdot}\delta_{y\cdot} g_{0,0} + f_{0,N_y}\delta_{x\cdot}\delta_{y\cdot} g_{0,N_y} + f_{N_x,0}\delta_{x\cdot}\delta_{y\cdot} g_{N_x,0} - f_{N_x,N_y}\delta_{x\cdot}\delta_{y\cdot} g_{N_x,N_y}
\end{aligned} \tag{16e}
$$

$$
\begin{aligned}
\langle \delta_{yy} f, \delta_{xx} g \rangle'_{\mathbb{U}^2_{N_x,N_y}} &= \langle \delta_{x+}\delta_{y+} f, \delta_{x+}\delta_{y+} g \rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \{\delta_{y\cdot} f, \delta_{xx} g\}'_{(\mathbb{U}_{N_x},0)} - \{f, \delta_{x\cdot}\delta_{yy} g\}'_{\left(0,\mathbb{U}_{N_y}\right)} \\
&\quad + \{\delta_{y\cdot} f, \delta_{xx} g\}'_{(\mathbb{U}_{N_x},N_y)} + \{f, \delta_{x\cdot}\delta_{yy} g\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} \\
&\quad - f_{0,0}\delta_{x\cdot}\delta_{y\cdot} g_{0,0} + f_{0,N_y}\delta_{x\cdot}\delta_{y\cdot} g_{0,N_y} + f_{N_x,0}\delta_{x\cdot}\delta_{y\cdot} g_{N_x,0} - f_{N_x,N_y}\delta_{x\cdot}\delta_{y\cdot} g_{N_x,N_y}
\end{aligned} \tag{16f}
$$

1.5 Some bounds

The following relations between norms of grid function under spatial difference operations to norms of the grid function themselves are of use in the energetic analysis of finite difference schemes.

$$
\begin{aligned}
\|\delta_{x+} u\|^2_{\underline{\mathbb{U}}} = \sum_{l=0}^{N-1} h \left(\delta_{x+} u_l\right)^2 = \sum_{l=0}^{N-1} \frac{1}{h} \left(u_{l+1} - u_l\right)^2 &= \sum_{l=0}^{N-1} \frac{1}{h} \left(u_{l+1}^2 + u_l^2 - 2u_{l+1}u_l\right) \\
&\leq \sum_{l=0}^{N-1} \frac{2}{h} \left(u_{l+1}^2 + u_l^2\right) \\
&= \sum_{l=1}^{N-1} \frac{4}{h} u_l^2 + \frac{2}{h} u_0^2 + \frac{2}{h} u_N^2 \\
&= \left(\frac{2}{h} \|u\|'_{\mathbb{U}}\right)^2 \leq \frac{4}{h^2} \|u\|^2_{\mathbb{U}}
\end{aligned} \tag{17a}
$$

$$\begin{aligned}
\|\delta_{x-}u\|_{\overline{\mathbb{U}}}^2 = \sum_{l=1}^{N} h\left(\delta_{x-}u_l\right)^2 = \sum_{l=1}^{N} \frac{1}{h}\left(u_l - u_{l-1}\right)^2 &= \sum_{l=1}^{N} \frac{1}{h}\left(u_l^2 + u_{l-1}^2 - 2u_l u_{l-1}\right) \\
&\leq \sum_{l=1}^{N} \frac{2}{h}\left(u_l^2 + u_{l-1}^2\right) \\
&= \sum_{l=1}^{N-1} \frac{4}{h} u_l^2 + \frac{2}{h}u_0^2 + \frac{2}{h}u_N^2 \\
&= \left(\frac{2}{h}\|u\|_{\mathbb{U}}'\right)^2 \leq \frac{4}{h^2}\|u\|_{\mathbb{U}}^2
\end{aligned} \tag{17b}$$

$$\begin{aligned}
\|\delta_{xx}u\|_{\underline{\mathbb{U}}}^2 = \|\delta_{x+}\delta_{x-}u\|_{\underline{\mathbb{U}}}^2 = \sum_{l=1}^{N-1} h\left(\delta_{x+}\delta_{x-}u_l\right)^2 &= \sum_{l=1}^{N-1} \frac{1}{h}\left[\left(\delta_{x+}u_l\right)^2 + \left(\delta_{x+}u_{l-1}\right)^2 - 2\delta_{x+}u_l\left(\delta_{x+}u_{l-1}\right)\right] \\
&\leq \sum_{l=1}^{N-1} \frac{2}{h}\left[\left(\delta_{x+}u_l\right)^2 + \left(\delta_{x+}u_{l-1}\right)^2\right] \\
&= \sum_{l=1}^{N-1} \frac{4}{h}\left(\delta_{x+}u_l\right)^2 + \frac{2}{h}\left(\delta_{x+}u_0\right)^2 - \frac{2}{h}\left(\delta_{x-}u_N\right)^2 \\
&\leq \frac{4}{h^2}\sum_{l=0}^{N-1} h\left(\delta_{x+}u_l\right)^2 \overset{(17a)}{\leq} \left(\frac{4}{h^2}\|u\|_{\mathbb{U}}'\right)^2 \leq \frac{16}{h^4}\|u\|_{\mathbb{U}}^2
\end{aligned} \tag{17c}$$

Starting from the following 3 relations:

$$\begin{aligned}
\sum_{l=0}^{N_x-1} h_x\left(\delta_{x+}u_{l,m}\right)^2 &\leq \sum_{l=1}^{N_x-1} h_x\left(\delta_{x+}u_{l,m}\right)^2 + \frac{h_x}{2}\left(\delta_{x+}u_{0,m}\right)^2 = \frac{1}{h_x}\sum_{l=1}^{N_x-1}\left(u_{l+1,m} - u_{l,m}\right)^2 + \frac{1}{2h_x}\left(u_{1,m} - u_{0,m}\right)^2 \\
&\leq \frac{2}{h_x^2}\left[\sum_{l=1}^{N_x-1} h_x u_{l+1,m}^2 + \sum_{l=1}^{N_x-1} h_x u_{l,m}^2\right] + \frac{1}{h_x}\left(u_{1,m}^2 + u_{0,m}^2\right) \\
&= \frac{4}{h_x^2}\sum_{l=1}^{N_x-1} h_x u_{l,m}^2 + \frac{2}{h_x}u_{N_x,m}^2 - \frac{1}{h_x}u_{1,m}^2 + \frac{1}{h_x}u_{0,m}^2 \\
&\leq \frac{4}{h_x^2}\sum_{l=1}^{N_x-1} h_x u_{l,m}^2 + \frac{2}{h_x}u_{N_x,m}^2 + \frac{2}{h_x}u_{0,m}^2 \\
\sum_{l=0}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y\left(\delta_{x+}u_{l,m}\right)^2 &\leq \frac{4}{h_x^2}\left[\sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y u_{l,m}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{N_x,m}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{0,m}^2\right]
\end{aligned}$$

$$\frac{h_x h_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,0}\right)^2 \overset{(17a)}{\leq} \frac{4}{h_x^2}\left[\frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,0}^2 + \frac{h_x h_y}{4}u_{0,0}^2 + \frac{h_x h_y}{4}u_{N_x,0}^2\right]$$

$$\frac{h_x h_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,N_y}\right)^2 \overset{(17a)}{\leq} \frac{4}{h_x^2}\left[\frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,N_y}^2 + \frac{h_x h_y}{4}u_{0,N_y}^2 + \frac{h_x h_y}{4}u_{N_x,N_y}^2\right]$$

these 3 inequalities may be combined to give

$$\sum_{l=0}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y \left(\delta_{x+}u_{l,m}\right)^2 + \frac{h_x h_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,0}\right)^2 + \frac{h_x h_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,N_y}\right)^2$$
$$\leq \frac{4}{h_x^2}\left[\sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y u_{l,m}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{N_x,m}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{0,m}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,0}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,N_y}^2\right]$$
$$+ \frac{4}{h_x^2}\left(\frac{h_x h_y}{4}u_{0,0}^2 + \frac{h_x h_y}{4}u_{N_x,0}^2 + \frac{h_x h_y}{4}u_{0,N_y}^2 + \frac{h_x h_y}{4}u_{N_x,N_y}^2\right)$$

or, written more succinctly

$$\|\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{h_y}{2}\|\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},0\right)} + \frac{h_y}{2}\|\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},N_y\right)} \leq \left(\frac{2}{h_x}\|u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 \leq \frac{4}{h_x^2}\|u\|^2_{\mathbb{U}^2_{N_x,N_y}} \tag{18a}$$

$$\|\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{h_x}{2}\|\delta_{y+}u\|^2_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} + \frac{h_x}{2}\|\delta_{y+}u\|^2_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)} \leq \left(\frac{2}{h_y}\|u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 \leq \frac{4}{h_y^2}\|u\|^2_{\mathbb{U}^2_{N_x,N_y}} \tag{18b}$$

Using the algebraic identity (resulting from expanding the expression on the left-hand side)

$$(a-2b+c)^2 = 4a^2 + 8b^2 + 4c^2 - 2(a+b)^2 - 2(b+c)^2 - (a-c)^2$$

for any real numbers a, b and c we may write

$$\begin{aligned}
\sum_{l=0}^{N_x} h_x\left(\delta_{xx}f_{l,m}\right)^2 &= \frac{1}{h_x^2}\sum_{l=0}^{N_x}\left(f_{l+1,m} - 2f_{l,m} + f_{l-1,m}\right)^2 \\
&= \frac{4}{h_x^3}\sum_{l=0}^{N_x} f_{l+1,m}^2 + \frac{8}{h_x^3}\sum_{l=0}^{N_x} f_{l,m} + \frac{4}{h_x^3}\sum_{l=0}^{N_x} f_{l-1,m} \\
&\quad - \frac{2}{h_x^3}\sum_{l=0}^{N_x}\left(f_{l+1,m}+f_{l,m}\right)^2 - \frac{2}{h_x^3}\sum_{l=0}^{N_x}\left(f_{l,m}+f_{l-1,m}\right)^2 - \frac{1}{h_x^3}\sum_{l=0}^{N_x}\left(f_{l+1,m}-f_{l-1,m}\right)^2 \\
&= \frac{16}{h_x^3}\sum_{l=0}^{N_x} f_{l,m}^2 + \frac{4}{h_x^3}\left(f_{-1,m}^2 - f_{0,m}^2\right) + \frac{4}{h_x^3}\left(f_{N_x+1,m}^2 - f_{N_x,m}^2\right) \\
&\quad - \frac{2}{h_x^3}\sum_{l=0}^{N_x}\left(f_{l+1,m}+f_{l,m}\right)^2 - \frac{2}{h_x^3}\sum_{l=0}^{N_x}\left(f_{l,m}+f_{l-1,m}\right)^2 - \frac{1}{h_x^3}\sum_{l=0}^{N_x}\left(f_{l+1,m}-f_{l-1,m}\right)^2
\end{aligned}$$

$$\begin{aligned}
4f_{-1,m}^2 - 4f_{0,m}^2 &= -2\left(f_{0,m}-f_{-1,m}\right)\left(f_{1,m}-2f_{0,m}+f_{-1,m}\right) \\
&\quad + 2\left(f_{0,m}+f_{-1,m}\right)^2 + \left(f_{1,m}-f_{-1,m}\right)^2 - 8f_{0,m}^2 - \left(f_{0,m}-f_{-1,m}\right)^2 - \left(f_{1,m}-f_{0,m}\right)^2 \\
&\leq -2\left(f_{0,m}-f_{-1,m}\right)\left(f_{1,m}-2f_{0,m}+f_{-1,m}\right) + 2\left(f_{0,m}+f_{-1,m}\right)^2 + \left(f_{1,m}-f_{-1,m}\right)^2 - 8f_{0,m}^2
\end{aligned}$$

$$\begin{aligned}
4f_{N_x+1,m}^2 - 4f_{N_x,m}^2 &= 2\left(f_{N_x+1,m}-f_{N_x,m}\right)\left(f_{N_x+1,m}-2f_{N_x,m}+f_{N_x-1,m}\right) + 2\left(f_{N_x+1,m}+f_{N_x,m}\right)^2 \\
&\quad + \left(f_{N_x+1,m}-f_{N_x-1,m}\right)^2 - 8f_{N_x,m}^2 - \left(f_{N_x+1,m}-f_{N_x,m}\right)^2 - \left(f_{N_x,m}-f_{N_x-1,m}\right)^2 \\
&\leq 2\left(f_{N_x+1,m}-f_{N_x,m}\right)\left(f_{N_x+1,m}-2f_{N_x,m}+f_{N_x-1,m}\right) + 2\left(f_{N_x+1,m}+f_{N_x,m}\right)^2 \\
&\quad + \left(f_{N_x+1,m}-f_{N_x-1,m}\right)^2 - 8f_{N_x,m}^2
\end{aligned}$$

these 3 relations may be combined to give

$$\sum_{l=0}^{N_x} h_x \left(\delta_{xx} f_{l,m}\right)^2 \leq \frac{16}{h_x^4} \sum_{l=0}^{N_x} h_x f_{l,m}^2 - \frac{8}{h_x^3} f_{0,m}^2 - \frac{8}{h_x^3} f_{N_x,m}^2 - 2\left(\delta_{x-} f_{0,m}\right) \delta_{xx} f_{0,m} + 2\left(\delta_{x+} f_{N_x,m}\right) \delta_{xx} f_{N_x,m}$$
$$\leq \frac{16}{h_x^4} \left(\sum_{l=1}^{N_x-1} h_x f_{l,m}^2 + \frac{h_x}{2} f_{0,m}^2 + \frac{h_x}{2} f_{N_x,m}^2 \right) - 2\left(\delta_{x-} f_{0,m}\right) \delta_{xx} f_{0,m} + 2\left(\delta_{x+} f_{N_x,m}\right) \delta_{xx} f_{N_x,m}$$

$$\sum_{l=1}^{N_x-1} h_x \left(\delta_{xx} f_{l,m}\right)^2 + \frac{h_x}{2} \left(\delta_{xx} f_{0,m}\right)^2 + \frac{h_x}{2} \left(\delta_{xx} f_{N_x,m}\right)^2 \leq \frac{16}{h_x^4} \left(\sum_{l=1}^{N_x-1} h_x f_{l,m}^2 + \frac{h_x}{2} f_{0,m}^2 + \frac{h_x}{2} f_{N_x,m}^2 \right)$$
$$- 2\left(\delta_{x\cdot} f_{0,m}\right) \delta_{xx} f_{0,m} + 2\left(\delta_{x\cdot} f_{N_x,m}\right) \delta_{xx} f_{N_x,m}$$

Summing over m then gives

$$\sum_{l=1}^{N_x-1} \sum_{m=1}^{N_y-1} h_x h_y \left(\delta_{xx} f_{l,m}\right)^2 + \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} \left(\delta_{xx} f_{0,m}\right)^2 + \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} \left(\delta_{xx} f_{N_x,m}\right)^2$$
$$\leq \frac{16}{h_x^4} \left(\sum_{l=1}^{N_x-1} \sum_{m=1}^{N_y-1} h_x h_y f_{l,m}^2 + \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} f_{0,m}^2 + \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} f_{N_x,m}^2 \right)$$
$$- 2 \sum_{m=1}^{N_y-1} h_y \left(\delta_{x\cdot} f_{0,m}\right) \delta_{xx} f_{0,m} + 2 \sum_{m=1}^{N_y-1} h_y \left(\delta_{x\cdot} f_{N_x,m}\right) \delta_{xx} f_{N_x,m}$$

Adding the two missing sums running over $l = 0$ to $l = N_x$ on the left-hand side and once more utilizing the combination of the 3 first relations for each associated sum on the right-hand side then finally results

$$\sum_{l=1}^{N_x-1} \sum_{m=1}^{N_y-1} h_x h_y \left(\delta_{xx} f_{l,m}\right)^2$$
$$+ \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} \left(\delta_{xx} f_{0,m}\right)^2 + \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} \left(\delta_{xx} f_{N_x,m}\right)^2 + \frac{h_x h_y}{2} \sum_{l=0}^{N_x} \left(\delta_{xx} f_{l,0}\right)^2 + \frac{h_x h_y}{2} \sum_{l=0}^{N_x} \left(\delta_{xx} f_{l,N_y}\right)^2$$
$$\leq \frac{16}{h_x^4} \left(\sum_{l=1}^{N_x-1} \sum_{m=1}^{N_y-1} h_x h_y f_{l,m}^2 + \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} f_{0,m}^2 + \frac{h_x h_y}{2} \sum_{m=1}^{N_y-1} f_{N_x,m}^2 + \frac{h_x h_y}{2} \sum_{l=1}^{N_x-1} f_{l,0}^2 + \frac{h_x h_y}{2} \sum_{l=1}^{N_x-1} f_{l,N_y}^2 \right)$$
$$+ \frac{16}{h_x^4} \left(\frac{h_x h_y}{4} f_{0,0}^2 + \frac{h_x h_y}{4} f_{N_x,0}^2 + \frac{h_x h_y}{4} f_{0,N_y}^2 + \frac{h_x h_y}{4} f_{N_x,N_y}^2 \right)$$
$$- 2 \sum_{m=1}^{N_y-1} h_y \left(\delta_{x\cdot} f_{0,m}\right) \delta_{xx} f_{0,m} + 2 \sum_{m=1}^{N_y-1} h_y \left(\delta_{x\cdot} f_{N_x,m}\right) \delta_{xx} f_{N_x,m}$$
$$- h_y \left(\delta_{x-} f_{0,0}\right) \delta_{xx} f_{0,0} + h_y \left(\delta_{x-} f_{N_x,0}\right) \delta_{xx} f_{N_x,0} - h_y \left(\delta_{x-} f_{0,N_y}\right) \delta_{xx} f_{0,N_y} + h_y \left(\delta_{x-} f_{N_x,N_y}\right) \delta_{xx} f_{N_x,N_y}$$

$$
\begin{aligned}
&\sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y \left(\delta_{xx} f_{l,m}\right)^2 \\
&\quad + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1}\left(\delta_{xx} f_{0,m}\right)^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1}\left(\delta_{xx} f_{N_x,m}\right)^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1}\left(\delta_{xx} f_{l,0}\right)^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1}\left(\delta_{xx} f_{l,N_y}\right)^2 \\
&\qquad + \frac{h_x h_y}{4}\left(\delta_{xx} f_{0,0}\right)^2 + \frac{h_x h_y}{4}\left(\delta_{xx} f_{N_x,0}\right)^2 + \frac{h_x h_y}{4}\left(\delta_{xx} f_{0,N_y}\right)^2 + \frac{h_x h_y}{4}\left(\delta_{xx} f_{N_x,N_y}\right)^2 \\
&\le \frac{16}{h_x^4}\left(\sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y f_{l,m}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} f_{0,m}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} f_{N_x,m}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} f_{l,0}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} f_{l,N_y}^2\right) \\
&\qquad + \frac{16}{h_x^4}\left(\frac{h_x h_y}{4} f_{0,0}^2 + \frac{h_x h_y}{4} f_{N_x,0}^2 + \frac{h_x h_y}{4} f_{0,N_y}^2 + \frac{h_x h_y}{4} f_{N_x,N_y}^2\right) \\
&\qquad - 2\sum_{m=1}^{N_y-1} h_y \left(\delta_{x\cdot} f_{0,m}\right)\delta_{xx} f_{0,m} + 2\sum_{m=1}^{N_y-1} h_y \left(\delta_{x\cdot} f_{N_x,m}\right)\delta_{xx} f_{N_x,m} \\
&\quad - h_y \left(\delta_{x\cdot} f_{0,0}\right)\delta_{xx} f_{0,0} + h_y \left(\delta_{x\cdot} f_{N_x,0}\right)\delta_{xx} f_{N_x,0} - h_y \left(\delta_{x\cdot} f_{0,N_y}\right)\delta_{xx} f_{0,N_y} + h_y \left(\delta_{x\cdot} f_{N_x,N_y}\right)\delta_{xx} f_{N_x,N_y}
\end{aligned}
$$

or, written more succinctly

$$
\left(\|\delta_{xx} f\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 \le \left(\frac{4}{h_x^2}\|f\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 - 2\left\{\delta_{x\cdot} f, \delta_{xx} f\right\}'_{(0,\mathbb{U}_{N_y})} + 2\left\{\delta_{x\cdot} f, \delta_{xx} f\right\}'_{(N_x,\mathbb{U}_{N_y})} \tag{19a}
$$

$$
\left(\|\delta_{yy} f\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 \le \left(\frac{4}{h_y^2}\|f\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 - 2\left\{\delta_{y\cdot} f, \delta_{yy} f\right\}'_{(\mathbb{U}_{N_x},0)} + 2\left\{\delta_{y\cdot} f, \delta_{yy} f\right\}'_{(\mathbb{U}_{N_x},N_y)} \tag{19b}
$$

Starting from the following 4 relations:

$$
\sum_{l=0}^{N_x-1} h_x \left(\delta_{x+}\delta_{y+} u_{l,m}\right)^2 \overset{(17a)}{\le} \frac{4}{h_x^2}\left[\sum_{l=1}^{N_x-1} h_x \left(\delta_{y+} u_{l,m}\right)^2 + \frac{h_x}{2}\left(\delta_{y+} u_{0,m}\right)^2 + \frac{h_x}{2}\left(\delta_{y+} u_{N_x,m}\right)^2\right]
$$

$$
\sum_{l=0}^{N_x-1}\sum_{m=0}^{N_y-1} h_x h_y \left(\delta_{x+}\delta_{y+} u_{l,m}\right)^2 = \frac{4}{h_x^2}\left[\sum_{l=1}^{N_x-1}\sum_{m=0}^{N_y-1} h_x h_y \left(\delta_{y+} u_{l,m}\right)^2 + \frac{h_x h_y}{2}\sum_{m=0}^{N_y-1}\left(\delta_{y+} u_{0,m}\right)^2 + \frac{h_x h_y}{2}\sum_{m=0}^{N_y-1}\left(\delta_{y+} u_{N_x,m}\right)^2\right]
$$

$$
\sum_{l=1}^{N_x-1}\sum_{m=0}^{N_y-1} h_x h_y \left(\delta_{y+} u_{l,m}\right)^2 \overset{(17a)}{\le} \frac{4}{h_y^2}\left[\sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y u_{l,m}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,0}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,N_y}^2\right]
$$

$$
\frac{h_x h_y}{2}\sum_{m=0}^{N_y-1}\left(\delta_{y+} u_{0,m}\right)^2 \overset{(17a)}{\le} \frac{4}{h_y^2}\left[\frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{0,m}^2 + \frac{h_x h_y}{4} u_{0,0}^2 + \frac{h_x h_y}{4} u_{0,N_y}^2\right]
$$

$$
\frac{h_x h_y}{2}\sum_{m=0}^{N_y-1}\left(\delta_{y+} u_{N_x,m}\right)^2 \overset{(17a)}{\le} \frac{4}{h_y^2}\left[\frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{N_x,m}^2 + \frac{h_x h_y}{4} u_{N_x,0}^2 + \frac{h_x h_y}{4} u_{N_x,N_y}^2\right]
$$

these 4 inequalities may be combined to give

$$\sum_{l=0}^{N_x-1}\sum_{m=0}^{N_y-1} h_x h_y \left(\delta_{x+}\delta_{y+}u_{l,m}\right)^2$$
$$\leq \frac{16}{h_x^2 h_y^2}\left[\sum_{l=1}^{N_x-1}\sum_{m=1}^{N_y-1} h_x h_y u_{l,m}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,0}^2 + \frac{h_x h_y}{2}\sum_{l=1}^{N_x-1} u_{l,N_y}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{0,m}^2 + \frac{h_x h_y}{2}\sum_{m=1}^{N_y-1} u_{N_x,m}^2\right]$$
$$+ \frac{16}{h_x^2 h_y^2}\left(\frac{h_x h_y}{4}u_{0,0}^2 + \frac{h_x h_y}{4}u_{0,N_y}^2 + \frac{h_x h_y}{4}u_{N_x,0}^2 + \frac{h_x h_y}{4}u_{N_x,N_y}^2\right)$$

or, written more succinctly

$$\|\delta_{x+}\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} \leq \left(\frac{4}{h_x h_y}\|u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 \leq \frac{16}{h_x^2 h_y^2}\|u\|^2_{\mathbb{U}^2_{N_x,N_y}} \tag{20}$$

1.6 Matrix interpretation of difference operators

Assuming the domain $\mathcal{D} = \mathbb{Z}$ for now, so $l = -\infty, \ldots, \infty$, the grid function u_l may be arranged in an infinite column vector $\mathbf{u} = [\ldots, u_{-1}, u_0, u_1, \ldots]^\top$. This allows us to express the operators δ_{x-}, δ_{x+}, δ_{xx} and δ_{xxxx} as the matrices $h^{-1}\mathbf{D}^{(1)}_{x-}$, $h^{-1}\mathbf{D}^{(1)}_{x+}$, $h^{-2}\mathbf{D}^{(1)}_{xx}$ and $h^{-4}\mathbf{D}^{(1)}_{xxxx}$ as follows:

$$\mathbf{D}^{(1)}_{x-} = \begin{bmatrix} \ddots & & & & & 0 \\ \ddots & 1 & & & & \\ & -1 & 1 & & & \\ & & -1 & 1 & & \\ & & & -1 & 1 & \\ 0 & & & & \ddots & \ddots \end{bmatrix} \quad \mathbf{D}^{(1)}_{x+} = \begin{bmatrix} \ddots & \ddots & & & & 0 \\ & -1 & 1 & & & \\ & & -1 & 1 & & \\ & & & -1 & 1 & \\ & & & & -1 & \ddots \\ 0 & & & & & \ddots \end{bmatrix}$$

$$\mathbf{D}^{(1)}_{xx} = \begin{bmatrix} \ddots & \ddots & & & & 0 \\ \ddots & -2 & 1 & & & \\ & 1 & -2 & 1 & & \\ & & 1 & -2 & 1 & \\ & & & 1 & -2 & \ddots \\ 0 & & & & \ddots & \ddots \end{bmatrix} \quad \mathbf{D}^{(1)}_{xxxx} = \begin{bmatrix} \ddots & \ddots & \ddots & & & & 0 \\ \ddots & 6 & -4 & 1 & & & \\ \ddots & -4 & 6 & -4 & 1 & & \\ & 1 & -4 & 6 & -4 & 1 & \\ & & 1 & -4 & 6 & -4 & \ddots \\ & & & 1 & -4 & 6 & \ddots \\ 0 & & & & \ddots & \ddots & \ddots \end{bmatrix} \tag{21}$$

2 Finite difference scheme for linear stiff strings

An explicit finite difference scheme for a stiff string with frequency dependent and independent loss [Bil09] is given by

$$\delta_{tt}u = \gamma^2\delta_{xx}u - \kappa^2\delta_{xxxx}u - 2\sigma_0\delta_{t\cdot}u + 2\sigma_1\delta_{t-}\delta_{xx}u \tag{22}$$

2.1 Energy and boundary conditions

Taking a primed l_2 inner product of (22) with $\delta_{t\cdot}u$ over the interval $\{l \in \mathbb{U}_N \mid 0 \leq l \leq N\}$ and rearranging terms results in

$$\langle \delta_{t\cdot} u, \delta_{tt} u \rangle'_{\mathbb{U}_N} - \gamma^2 \langle \delta_{t\cdot} u, \delta_{xx} u \rangle'_{\mathbb{U}_N} + \kappa^2 \langle \delta_{t\cdot} u, \delta_{xxxx} u \rangle'_{\mathbb{U}_N} = -2\sigma_0 \left(\|\delta_{t\cdot} u\|'_{\mathbb{U}_N} \right)^2 + 2\sigma_1 \langle \delta_{t\cdot} u, \delta_{t-}\delta_{xx} u \rangle'_{\mathbb{U}_N}$$

After employing summation by parts (13f), (13i) one has

$$\langle \delta_{t\cdot} u, \delta_{tt} u \rangle'_{\mathbb{U}_N} + \gamma^2 \langle \delta_{t\cdot}\delta_{x+} u, \delta_{x+} u \rangle_{\underline{\mathbb{U}}_N} + \kappa^2 \langle \delta_{t\cdot}\delta_{xx} u, \delta_{xx} u \rangle_{\overline{\mathbb{U}}_N} = \mathfrak{b} - \mathfrak{q}_0$$

where the boundary terms denoted by $\mathfrak{b}$ are given by

$$\begin{aligned}\mathfrak{b} \triangleq & -\gamma^2 \left(\delta_{t\cdot} u_0\right) \delta_{x\cdot} u_0 + \gamma^2 \left(\delta_{t\cdot} u_N\right) \delta_{x\cdot} u_N \\ & + \kappa^2 \left(\delta_{t\cdot} u_0\right) \delta_{x\cdot}\delta_{xx} u_0 - \kappa^2 \left(\delta_{t\cdot} u_N\right) \delta_{x\cdot}\delta_{xx} u_N - \kappa^2 \left(\delta_{xx} u_0\right) \delta_{t\cdot}\delta_{x\cdot} u_0 + \kappa^2 \left(\delta_{xx} u_N\right) \delta_{t\cdot}\delta_{x\cdot} u_N \\ & - 2\sigma_1 \left(\delta_{t\cdot} u_0\right) \delta_{t-}\delta_{x\cdot} u_0 + 2\sigma_1 \left(\delta_{t\cdot} u_N\right) \delta_{t-}\delta_{x\cdot} u_N\end{aligned}$$

and the loss terms denoted by $\mathfrak{q}_0$ are given by

$$\mathfrak{q}_0 = 2\sigma_0 \left(\|\delta_{t\cdot} u\|'_{\mathbb{U}_N} \right)^2 + 2\sigma_1 \langle \delta_{t\cdot}\delta_{x+} u, \delta_{t-}\delta_{x+} u \rangle_{\underline{\mathbb{U}}_N}$$

The term involving frequency dependent loss may, using identity (3g), be written as

$$2\sigma_1 \langle \delta_{t\cdot}\delta_{x+} u, \delta_{t-}\delta_{x+} u \rangle_{\underline{\mathbb{U}}_N} = 2\sigma_1 \|\delta_{t\cdot}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} - \sigma_1 k \langle \delta_{t\cdot}\delta_{x+} u, \delta_{tt}\delta_{x+} u \rangle_{\underline{\mathbb{U}}_N}$$

Using (4a) and (4b) the left-hand side of the inner product together with the second frequency dependent loss term involving the forward difference may be written as the total difference

$$\delta_{t+} \left(\frac{1}{2} \left(\|\delta_{t-} u\|'_{\mathbb{U}_N} \right)^2 + \frac{\gamma^2}{2} \langle \delta_{x+} u, e_{t-}\delta_{x+} u \rangle_{\underline{\mathbb{U}}_N} + \frac{\kappa^2}{2} \langle \delta_{xx} u, e_{t-}\delta_{xx} u \rangle_{\overline{\mathbb{U}}_N} - \frac{\sigma_1 k}{2} \|\delta_{t-}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} \right) = \delta_{t+} \mathfrak{h}$$

so that

$$\delta_{t+} \mathfrak{h} = \mathfrak{b} - \mathfrak{q} \tag{23}$$

where

$$\mathfrak{t} = \frac{1}{2} \left(\|\delta_{t-} u\|'_{\mathbb{U}_N} \right)^2 \tag{24a}$$

$$\mathfrak{v} = \frac{\gamma^2}{2} \langle \delta_{x+} u, e_{t-}\delta_{x+} u \rangle_{\underline{\mathbb{U}}_N} + \frac{\kappa^2}{2} \langle \delta_{xx} u, e_{t-}\delta_{xx} u \rangle_{\overline{\mathbb{U}}_N} - \frac{\sigma_1 k}{2} \|\delta_{t-}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} \tag{24b}$$

$$\mathfrak{h} = \mathfrak{t} + \mathfrak{v} \tag{24c}$$

$$\mathfrak{q} = 2\sigma_0 \left(\|\delta_{t\cdot} u\|'_{\mathbb{U}_N} \right)^2 + 2\sigma_1 \|\delta_{t\cdot}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} \tag{24d}$$

are the kinetic, potential and numerical energy and loss terms respectively. *Note that for the explicit scheme* (22) *the potential energy includes a term related to the original frequency dependent loss term.*

To determine the conditions under which the numerical energy $\mathfrak{h}$ is positive definite, we can write, using (4c), (17a) and (17c), for the potential energy

$$\begin{aligned}\mathfrak{v} &= \frac{\gamma^2}{2} \langle \delta_{x+} u, e_{t-}\delta_{x+} u \rangle_{\underline{\mathbb{U}}_N} + \frac{\kappa^2}{2} \langle \delta_{xx} u, e_{t-}\delta_{xx} u \rangle_{\overline{\mathbb{U}}_N} - \frac{\sigma_1 k}{2} \|\delta_{t-}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} \\ &= \frac{\gamma^2}{2} \left(\|\mu_{t-}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} - \frac{k^2}{4} \|\delta_{t-}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} \right) + \frac{\kappa^2}{2} \left(\|\mu_{t-}\delta_{xx} u\|^2_{\overline{\mathbb{U}}_N} - \frac{k^2}{4} \|\delta_{t-}\delta_{xx} u\|^2_{\overline{\mathbb{U}}_N} \right) - \frac{\sigma_1 k}{2} \|\delta_{t-}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} \\ &\geq \frac{\gamma^2}{2} \left(\|\mu_{t-}\delta_{x+} u\|^2_{\underline{\mathbb{U}}_N} - \frac{k^2}{h^2} \left(\|\delta_{t-} u\|'_{\mathbb{U}_N} \right)^2 \right) + \frac{\kappa^2}{2} \left(\|\mu_{t-}\delta_{xx} u\|^2_{\overline{\mathbb{U}}_N} - \frac{4k^2}{h^4} \left(\|\delta_{t-} u\|'_{\mathbb{U}_N} \right)^2 \right) - \frac{2\sigma_1 k}{h^2} \left(\|\delta_{t-} u\|'_{\mathbb{U}_N} \right)^2\end{aligned}$$

Thus, we have for the total energy

$$\mathfrak{h} = \mathfrak{t} + \mathfrak{v} \geq \frac{1}{2}\left(1 - \lambda^2 - 4\mu^2 - \frac{4\sigma_1 k}{h^2}\right)\left(\|\delta_{t-}u\|'_{\mathbb{U}_N}\right)^2 + \frac{\gamma^2}{2}\|\mu_{t-}\delta_{x+}u\|^2_{\underline{\mathbb{U}}_N} + \frac{\kappa^2}{2}\|\mu_{t-}\delta_{xx}u\|^2_{\overline{\underline{\mathbb{U}}}_N}$$

where

$$\lambda = \frac{\gamma k}{h} \qquad\qquad \mu = \frac{\kappa k}{h^2} \tag{25}$$

This discrete energy is thus non-negative under the condition

$$1 - \lambda^2 - 4\mu^2 - \frac{4\sigma_1 k}{h^2} \geq 0$$

or

$$h \geq \sqrt{\frac{1}{2}\left(\gamma^2 k^2 + 4\sigma_1 k + \sqrt{\left(\gamma^2 k^2 + 4\sigma_1 k\right)^2 + 16\kappa^2 k^2}\right)} \tag{26}$$

The boundary terms denoted by $\mathfrak{b}$ vanish under the following choices of numerical boundary condition at the left and right endpoints of a domain:

$$\begin{aligned} u_0 = \delta_{x\cdot}u_0 &= 0 \\ u_N = \delta_{x\cdot}u_N &= 0 \end{aligned} \quad \text{clamped} \tag{27a}$$

$$\begin{aligned} u_0 = \delta_{xx}u_0 &= 0 \\ u_N = \delta_{xx}u_N &= 0 \end{aligned} \quad \text{simply supported} \tag{27b}$$

$$\begin{aligned} \delta_{xx}u_0 = \gamma^2\delta_{x\cdot}u_0 - \kappa^2\delta_{x\cdot}\delta_{xx}u_0 + 2\sigma_1\delta_{t-}\delta_{x\cdot}u_0 &= 0 \\ \delta_{xx}u_N = \gamma^2\delta_{x\cdot}u_0 - \kappa^2\delta_{x\cdot}\delta_{xx}u_N + 2\sigma_1\delta_{t-}\delta_{x\cdot}u_N &= 0 \end{aligned} \quad \text{free} \tag{27c}$$

2.2 Matrix form

If we define the column state vector $\mathbf{u}^n$ as

$$\mathbf{u}^n = [u_0^n, \ldots, u_N^n]^\top \tag{28}$$

where $N = \lfloor 1/h \rfloor$, scheme (22) may be rewritten in vector-matrix form as

$$\mathbf{u}^{n+1} = \mathbf{B}\mathbf{u}^n + \mathbf{C}\mathbf{u}^{n-1} \tag{29}$$

where the matrices $\mathbf{B}$ and $\mathbf{C}$ are defined as

$$\mathbf{B} = \frac{1}{1+\sigma_0 k}\left[2\mathbf{I} + \left(\frac{\gamma^2 k^2}{h^2} + \frac{2\sigma_1 k}{h^2}\right)\mathbf{D}_{xx}^{(1)} - \frac{\kappa^2 k^2}{h^4}\mathbf{D}_{xxxx}^{(1)}\right] \tag{30a}$$

$$\mathbf{C} = -\frac{1}{1+\sigma_0 k}\left[(1-\sigma_0 k)\,\mathbf{I} + \frac{2\sigma_1 k}{h^2}\mathbf{D}_{xx}^{(1)}\right] \tag{30b}$$

and the matrices $\mathbf{D}_{xx}^{(1)}$ and $\mathbf{D}_{xxxx}^{(1)}$ are the finite versions of the ones defined by (22) with the relevant boundary conditions applied. Under clamped (27a) or simply supported (27b) conditions at both ends they take the form

$$\mathbf{D}_{xx}^{(1)} = \underbrace{\begin{bmatrix} 0 & 0 & & & & & 0 \\ 0 & -2 & 1 & & & & \\ & 1 & -2 & 1 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 1 & -2 & 1 & \\ & & & & 1 & -2 & 0 \\ 0 & & & & & 0 & 0 \end{bmatrix}}_{\text{clamped or simply supported}}$$

$$\mathbf{D}_{xxxx}^{(1)} = \underbrace{\begin{bmatrix} 0 & 0 & & & & & & & 0 \\ 0 & 7 & -4 & 1 & & & & & \\ 0 & -4 & 6 & -4 & 1 & & & & \\ & 1 & -4 & 6 & -4 & 1 & & & \\ & & \ddots & \ddots & \ddots & \ddots & \ddots & & \\ & & & 1 & -4 & 6 & -4 & 1 & \\ & & & & 1 & -4 & 6 & -4 & 0 \\ & & & & & 1 & -4 & 7 & 0 \\ 0 & & & & & & & 0 & 0 \end{bmatrix}}_{\text{clamped}}$$

$$\mathbf{D}_{xxxx}^{(1)} = \underbrace{\begin{bmatrix} 0 & 0 & & & & & & & 0 \\ 0 & 5 & -4 & 1 & & & & & \\ 0 & -4 & 6 & -4 & 1 & & & & \\ & 1 & -4 & 6 & -4 & 1 & & & \\ & & \ddots & \ddots & \ddots & \ddots & \ddots & & \\ & & & 1 & -4 & 6 & -4 & 1 & \\ & & & & 1 & -4 & 6 & -4 & 0 \\ & & & & & 1 & -4 & 5 & 0 \\ 0 & & & & & & & 0 & 0 \end{bmatrix}}_{\text{simply supported}}$$

Free conditions are more complicated. Assuming free conditions at the left endpoint, then using the first condition we find

$$u_{-1}^n = 2u_0^n - u_1^n$$

Substituting this in the second condition we find

$$u_{-2}^n = 2\beta\left(u_0^n - u_1^n\right) + u_2^n - 2\alpha\left(u_0^{n-1} - u_1^{n-1}\right)$$

where

$$\alpha = \frac{2\sigma_1 h^2}{\kappa^2 k} \qquad\qquad \beta = 2 + \frac{\gamma^2 h^2}{\kappa^2} + \alpha$$

Now we can express the operator δ_{xxxx} acting on the grid function u_l^n at grid points $l = 0$ and $l = 1$ fully in terms of grid points lying in the domain interior:

$$\begin{aligned} h^4\delta_{xxxx}u_0^n &= 2\left(\left(\beta - 1\right)u_0^n - \beta u_1^n + u_2^n - \alpha\left(u_0^{n-1} - u_1^{n-1}\right)\right) \\ h^4\delta_{xxxx}u_1^n &= -2u_0^n + 5u_1^n - 4u_2^n + u_3^n \end{aligned}$$

As such, the matrix $\mathbf{D}_{xx}^{(1)}$ under free boundary conditions at both ends takes the form

$$\mathbf{D}_{xx}^{(1)} = \underbrace{\begin{bmatrix} 0 & 0 & & & 0 \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ 0 & & & 0 & 0 \end{bmatrix}}_{\text{free}}$$

Expressing the matrix $\mathbf{D}_{xxxx}^{(1)}$ under free boundary conditions is less straightforward though, due to the dependence of the grid function u_l at the point $l = 0$ (and / or $l = N$) on values computed at the previous time instant $n-1$. One way to express this is by splitting $\mathbf{D}_{xxxx}$ into two matrices, one of which is acting on $\mathbf{u}^n$ and the other one on $\mathbf{u}^{n-1}$. Then we have, assuming free boundary conditions at both ends:

$$\begin{aligned}\mathbf{D}_{xxxx}^{(1)} &= \tilde{\mathbf{D}}_{xxxx}^{(1,n)} + \tilde{\mathbf{D}}_{xxxx}^{(1,n-1)} \\ &= \underbrace{\begin{bmatrix} 2(\beta-1) & -2\beta & 2 & & & & 0 \\ -2 & 5 & -4 & 1 & & & \\ 1 & -4 & 6 & -4 & 1 & & \\ & \ddots & \ddots & \ddots & \ddots & \ddots & \\ & & 1 & -4 & 6 & -4 & 1 \\ & & & 1 & -4 & 5 & -2 \\ 0 & & & & 2 & -2\beta & 2(\beta-1) \end{bmatrix}}_{\text{acting on } \mathbf{u}^n} + \underbrace{\begin{bmatrix} -2\alpha & 2\alpha & & & 0 \\ 0 & 0 & 0 & & \\ & \ddots & \ddots & \ddots & \\ & & 0 & 0 & 0 \\ 0 & & & 2\alpha & -2\alpha \end{bmatrix}}_{\text{acting on } \mathbf{u}^{n-1}}\end{aligned}$$

Hence, in the case of free boundary conditions at both ends equation (30a) and (30b) become

$$\begin{aligned}\mathbf{B} &= \frac{1}{1+\sigma_0 k}\left[2\mathbf{I} + \left(\frac{\gamma^2 k^2}{h^2} + \frac{2\sigma_1 k}{h^2}\right)\mathbf{D}_{xx}^{(1)} - \frac{\kappa^2 k^2}{h^4}\tilde{\mathbf{D}}_{xxxx}^{(1,n)}\right] \\ \mathbf{C} &= -\frac{1}{1+\sigma_0 k}\left[(1-\sigma_0 k)\,\mathbf{I} + \frac{2\sigma_1 k}{h^2}\mathbf{D}_{xx}^{(1)} + \frac{\kappa^2 k^2}{h^4}\tilde{\mathbf{D}}_{xxxx}^{(1,n-1)}\right]\end{aligned}$$

3 Finite difference scheme for linear stiff membranes

An explicit finite difference scheme for a stiff membrane with frequency dependent and independent loss [Bil09] is given by

$$\delta_{tt} u = \gamma^2 \delta_{\Delta\boxplus} u - \kappa^2 \delta_{\Delta\boxplus,\Delta\boxplus} u - 2\sigma_0 \delta_{t\cdot} u + 2\sigma_1 \delta_{t-} \delta_{\Delta\boxplus} u \tag{31}$$

3.1 Energy and boundary conditions

Taking a primed l_2 inner product of (31) with $\delta_{t\cdot} u$ over the region $\{(l,m) \in \mathbb{U}_{N_x,N_y} \mid 0 \le l \le N_x, 0 \le m \le N_y\}$ and rearranging terms results in

$$\begin{aligned}&\langle \delta_{t\cdot} u, \delta_{tt} u\rangle'_{\mathbb{U}^2_{N_x,N_y}} - \gamma^2 \langle \delta_{t\cdot} u, \delta_{\Delta\boxplus} u\rangle'_{\mathbb{U}^2_{N_x,N_y}} \\ &\qquad + \kappa^2 \langle \delta_{t\cdot} u, \delta_{\Delta\boxplus,\Delta\boxplus} u\rangle'_{\mathbb{U}^2_{N_x,N_y}} = -2\sigma_0 \left(\|\delta_{t\cdot} u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 + 2\sigma_1 \langle \delta_{t\cdot} u, \delta_{t-}\delta_{\Delta\boxplus} u\rangle'_{\mathbb{U}^2_{N_x,N_y}}\end{aligned}$$

After employing summation by parts (14b), (15c) one has

$$\begin{aligned}&\langle \delta_{t\cdot} u, \delta_{tt} u\rangle'_{\mathbb{U}^2_{N_x,N_y}} \\ &\quad + \gamma^2 \langle \delta_{t\cdot}\delta_{x+} u, \delta_{x+} u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{\gamma^2 h_y}{2}\left\{\delta_{t\cdot}\delta_{x+} u, \delta_{x+} u\right\}_{(\underline{\mathbb{U}}_{N_x},0)} + \frac{\gamma^2 h_y}{2}\left\{\delta_{t\cdot}\delta_{x+} u, \delta_{x+} u\right\}_{(\underline{\mathbb{U}}_{N_x},N_y)} \\ &\quad + \gamma^2 \langle \delta_{t\cdot}\delta_{y+} u, \delta_{y+} u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{\gamma^2 h_x}{2}\left\{\delta_{t\cdot}\delta_{y+} u, \delta_{y+} u\right\}_{(0,\underline{\mathbb{U}}_{N_y})} + \frac{\gamma^2 h_x}{2}\left\{\delta_{t\cdot}\delta_{y+} u, \delta_{y+} u\right\}_{(N_x,\underline{\mathbb{U}}_{N_y})} \\ &\qquad + \kappa^2 \langle \delta_{t\cdot}\delta_{\Delta\boxplus} u, \delta_{\Delta\boxplus} u\rangle'_{\mathbb{U}^2_{N_x,N_y}} = \mathfrak{b}_0 - \mathfrak{q}_0\end{aligned}$$

where the boundary terms denoted by $\mathfrak{b}_0$ are given by

$$\begin{aligned}
\mathfrak{b}_0 = &-\gamma^2 \left\{\delta_{t.}u, \delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},0)} - \gamma^2 \left\{\delta_{t.}u, \delta_{x.}u\right\}'_{\left(0,\mathbb{U}_{N_y}\right)} + \gamma^2 \left\{\delta_{t.}u, \delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},N_y)} + \gamma^2 \left\{\delta_{t.}u, \delta_{x.}u\right\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} \\
&+ \kappa^2 \left\{\delta_{t.}u, \delta_{y.}\delta_{\Delta\boxplus}u\right\}'_{(\mathbb{U}_{N_x},0)} - \kappa^2 \left\{\delta_{t.}\delta_{y.}u, \delta_{\Delta\boxplus}u\right\}'_{(\mathbb{U}_{N_x},0)} + \kappa^2 \left\{\delta_{t.}u, \delta_{x.}\delta_{\Delta\boxplus}u\right\}'_{\left(0,\mathbb{U}_{N_y}\right)} - \kappa^2 \left\{\delta_{t.}\delta_{x.}u, \delta_{\Delta\boxplus}u\right\}'_{\left(0,\mathbb{U}_{N_y}\right)} \\
&- \kappa^2 \left\{\delta_{t.}u, \delta_{y.}\delta_{\Delta\boxplus}u\right\}'_{(\mathbb{U}_{N_x},N_y)} + \kappa^2 \left\{\delta_{t.}\delta_{y.}u, \delta_{\Delta\boxplus}u\right\}'_{(\mathbb{U}_{N_x},N_y)} - \kappa^2 \left\{\delta_{t.}u, \delta_{x.}\delta_{\Delta\boxplus}u\right\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} + \kappa^2 \left\{\delta_{t.}\delta_{x.}u, \delta_{\Delta\boxplus}u\right\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} \\
&- 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},0)} - 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{x.}u\right\}'_{\left(0,\mathbb{U}_{N_y}\right)} + 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},N_y)} + 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{x.}u\right\}'_{\left(N_x,\mathbb{U}_{N_y}\right)}
\end{aligned}$$

and the loss terms denoted by $\mathfrak{q}_0$ are given by

$$\begin{aligned}
\mathfrak{q}_0 = &\, 2\sigma_0 \|\delta_{t.}u\|^2_{\mathbb{U}^2_{N_x,N_y}} \\
&+ 2\sigma_1 \langle \delta_{t.}\delta_{x+}u, \delta_{t-}\delta_{x+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \sigma_1 h_y \left\{\delta_{t.}\delta_{x+}u, \delta_{t-}\delta_{x+}u\right\}_{\left(\underline{\mathbb{U}}_{N_x},0\right)} + \sigma_1 h_y \left\{\delta_{t.}\delta_{x+}u, \delta_{t-}\delta_{x+}u\right\}_{\left(\underline{\mathbb{U}}_{N_x},N_y\right)} \\
&+ 2\sigma_1 \langle \delta_{t.}\delta_{y+}u, \delta_{t-}\delta_{y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \sigma_1 h_x \left\{\delta_{t.}\delta_{y+}u, \delta_{t-}\delta_{y+}u\right\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} + \sigma_1 h_x \left\{\delta_{t.}\delta_{y+}u, \delta_{t-}\delta_{y+}u\right\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}
\end{aligned}$$

The terms involving frequency dependent loss may, using identity (3g), be written as

$$\begin{aligned}
&2\sigma_1 \langle \delta_{t.}\delta_{x+}u, \delta_{t-}\delta_{x+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \sigma_1 h_y \left\{\delta_{t.}\delta_{x+}u, \delta_{t-}\delta_{x+}u\right\}_{\left(\underline{\mathbb{U}}_{N_x},0\right)} + \sigma_1 h_y \left\{\delta_{t.}\delta_{x+}u, \delta_{t-}\delta_{x+}u\right\}_{\left(\underline{\mathbb{U}}_{N_x},N_y\right)} \\
&\qquad = 2\sigma_1 \|\delta_{t.}\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \sigma_1 h_y \|\delta_{t.}\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},0\right)} + \sigma_1 h_y \|\delta_{t.}\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},N_y\right)} \\
&\quad - \sigma_1 k \langle \delta_{t.}\delta_{x+}u, \delta_{tt}\delta_{x+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} + \frac{\sigma_1 h_y k}{2} \left\{\delta_{t.}\delta_{x+}u, \delta_{tt}\delta_{x+}u\right\}_{\left(\underline{\mathbb{U}}_{N_x},0\right)} - \frac{\sigma_1 h_y k}{2} \left\{\delta_{t.}\delta_{x+}u, \delta_{tt}\delta_{x+}u\right\}_{\left(\underline{\mathbb{U}}_{N_x},N_y\right)}
\end{aligned}$$

and

$$\begin{aligned}
&2\sigma_1 \langle \delta_{t.}\delta_{y+}u, \delta_{t-}\delta_{y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \sigma_1 h_x \left\{\delta_{t.}\delta_{y+}u, \delta_{t-}\delta_{y+}u\right\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} + \sigma_1 h_x \left\{\delta_{t.}\delta_{y+}u, \delta_{t-}\delta_{y+}u\right\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)} \\
&\qquad = 2\sigma_1 \|\delta_{t.}\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \sigma_1 h_x \|\delta_{t.}\delta_{y+}u\|^2_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} + \sigma_1 h_x \|\delta_{t.}\delta_{y+}u\|^2_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)} \\
&\quad - \sigma_1 k \langle \delta_{t.}\delta_{y+}u, \delta_{tt}\delta_{y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} + \frac{\sigma_1 h_x k}{2} \left\{\delta_{t.}\delta_{y+}u, \delta_{tt}\delta_{y+}u\right\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} - \frac{\sigma_1 h_x k}{2} \left\{\delta_{t.}\delta_{y+}u, \delta_{tt}\delta_{y+}u\right\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}
\end{aligned}$$

Note that for the first three terms on the right-hand side it is true that

$$\begin{aligned}
&2\sigma_1 \|\delta_{t.}\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \sigma_1 h_y \|\delta_{t.}\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},0\right)} + \sigma_1 h_y \|\delta_{t.}\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},N_y\right)} \\
&\quad = 2\sigma_1 \sum_{l=0}^{N_x-1} \sum_{m=0}^{N_y-1} h_x h_y \left(\delta_{t.}\delta_{x+}u_{l,m}\right)^2 - \sigma_1 \sum_{l=0}^{N_x-1} h_x h_y \left(\delta_{t.}\delta_{x+}u_{l,0}\right)^2 + \sigma_1 \sum_{l=0}^{N_x-1} h_x h_y \left(\delta_{t.}\delta_{x+}u_{l,N_y}\right)^2 \\
&\quad = 2\sigma_1 \sum_{l=0}^{N_x-1} \sum_{m=1}^{N_y-1} h_x h_y \left(\delta_{t.}\delta_{x+}u_{l,m}\right)^2 + \sigma_1 \sum_{l=0}^{N_x-1} h_x h_y \left(\delta_{t.}\delta_{x+}u_{l,0}\right)^2 + \sigma_1 \sum_{l=0}^{N_x-1} h_x h_y \left(\delta_{t.}\delta_{x+}u_{l,N_y}\right)^2 \geq 0
\end{aligned}$$

The same holds for the terms of $\mathfrak{q}_0$ involving the forward differences in y of course. Using (4a) and (4b) the left-hand side of the inner product together with the last three frequency dependent loss terms for both x and y involving the forward difference may be written as the total difference

$$\delta_{t+}\left(\frac{1}{2}\|\delta_{t-}u\|^2_{\mathbb{U}^2_{N_x,N_y}}\right)+\delta_{t+}\left(\frac{\gamma^2}{2}\langle\delta_{x+}u,e_{t-}\delta_{x+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}}+\frac{\gamma^2}{2}\langle\delta_{y+}u,e_{t-}\delta_{y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}}\right)$$
$$+\delta_{t+}\left(\frac{\kappa^2}{2}\langle\delta_{\Delta\boxplus}u,e_{t-}\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}-\frac{\sigma_1 k}{2}\|\delta_{t-}\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}-\frac{\sigma_1 k}{2}\|\delta_{t-}\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}\right)$$
$$-\delta_{t+}\left(\frac{\gamma^2 h_y}{4}\left\{\delta_{x+}u,e_{t-}\delta_{x+}u\right\}_{(\underline{\mathbb{U}}_{N_x},0)}-\frac{\gamma^2 h_y}{4}\left\{\delta_{x+}u,e_{t-}\delta_{x+}u\right\}_{(\underline{\mathbb{U}}_{N_x},N_y)}\right)$$
$$-\delta_{t+}\left(\frac{\gamma^2 h_x}{4}\left\{\delta_{y+}u,e_{t-}\delta_{y+}u\right\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)}-\frac{\gamma^2 h_x}{4}\left\{\delta_{y+}u,e_{t-}\delta_{y+}u\right\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\right)$$
$$-\delta_{t+}\left(\frac{\sigma_1 h_y k}{4}\|\delta_{t-}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},0)}-\frac{\sigma_1 h_y k}{4}\|\delta_{t-}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},N_y)}\right)$$
$$-\delta_{t+}\left(\frac{\sigma_1 h_x k}{4}\|\delta_{t-}\delta_{y+}u\|^2_{\left(0,\underline{\mathbb{U}}_{N_y}\right)}-\frac{\sigma_1 h_x k}{4}\|\delta_{t-}\delta_{y+}u\|^2_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\right)=\delta_{t+}\mathfrak{h}_0$$

so that

$$\delta_{t+}\mathfrak{h}_0=\mathfrak{b}_0-\mathfrak{q} \tag{32}$$

where

$$\mathfrak{t}=\frac{1}{2}\|\delta_{t-}u\|^2_{\mathbb{U}^2_{N_x,N_y}} \tag{33a}$$

$$\begin{aligned}\mathfrak{v}_0=&\frac{\gamma^2}{2}\langle\delta_{x+}u,e_{t-}\delta_{x+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}}-\frac{\gamma^2 h_y}{4}\left\{\delta_{x+}u,e_{t-}\delta_{x+}u\right\}_{(\underline{\mathbb{U}}_{N_x},0)}+\frac{\gamma^2 h_y}{4}\left\{\delta_{x+}u,e_{t-}\delta_{x+}u\right\}_{(\underline{\mathbb{U}}_{N_x},N_y)}\\&+\frac{\gamma^2}{2}\langle\delta_{y+}u,e_{t-}\delta_{y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}}-\frac{\gamma^2 h_x}{4}\left\{\delta_{y+}u,e_{t-}\delta_{y+}u\right\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)}+\frac{\gamma^2 h_x}{4}\left\{\delta_{y+}u,e_{t-}\delta_{y+}u\right\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\\&-\frac{\sigma_1 k}{2}\|\delta_{t-}\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}+\frac{\sigma_1 h_y k}{4}\|\delta_{t-}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},0)}-\frac{\sigma_1 h_y k}{4}\|\delta_{t-}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},N_y)}\\&-\frac{\sigma_1 k}{2}\|\delta_{t-}\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}+\frac{\sigma_1 h_x k}{4}\|\delta_{t-}\delta_{y+}u\|^2_{\left(0,\underline{\mathbb{U}}_{N_y}\right)}-\frac{\sigma_1 h_x k}{4}\|\delta_{t-}\delta_{y+}u\|^2_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\end{aligned} \tag{33b}$$

$$\mathfrak{h}_0=\mathfrak{t}+\mathfrak{v}_0+\frac{\kappa^2}{2}\langle\delta_{\Delta\boxplus}u,e_{t-}\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} \tag{33c}$$

$$\begin{aligned}\mathfrak{q}=&2\sigma_0\|\delta_{t\cdot}u\|^2_{\mathbb{U}^2_{N_x,N_y}}\\&+2\sigma_1\|\delta_{t\cdot}\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}-\sigma_1 h_y\|\delta_{t\cdot}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},0)}+\sigma_1 h_y\|\delta_{t\cdot}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},N_y)}\\&+2\sigma_1\|\delta_{t\cdot}\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}-\sigma_1 h_x\|\delta_{t\cdot}\delta_{y+}u\|^2_{\left(0,\underline{\mathbb{U}}_{N_y}\right)}+\sigma_1 h_x\|\delta_{t\cdot}\delta_{y+}u\|^2_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\end{aligned} \tag{33d}$$

are the kinetic, potential and numerical energy and loss terms respectively. *Note that for the explicit scheme* (31) *the potential energy includes terms related to the original frequency dependent loss term. Furthermore, the potential energy also contains various boundary terms that will be of use later on when discussing the conditions under which the numerical energy remains non-negative.*

In contrast to the case of the stiff string with frequency dependent and independent loss it is not possible to extract free boundary conditions from $\mathfrak{b}_0$ directly[1]. Defining $v \triangleq \delta_{t\cdot}u$ one may write

[1]See section 12.1.2, page 334 and 335 of [Bil09] for more context.

$$\begin{aligned}\kappa^2\langle\delta_{\Delta\boxplus}v,\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}_{N_x,N_y}} &= \kappa^2\nu\langle\delta_{\Delta\boxplus}v,\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}_{N_x,N_y}} + \kappa^2(1-\nu)\langle\delta_{\Delta\boxplus}v,\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}_{N_x,N_y}}\\ &= \kappa^2\nu\langle\delta_{\Delta\boxplus}v,\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}_{N_x,N_y}} + \kappa^2(1-\nu)\left(\langle\delta_{xx}v,\delta_{xx}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \langle\delta_{yy}v,\delta_{yy}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}\right)\\ &\quad + \kappa^2(1-\nu)\left(\langle\delta_{xx}v,\delta_{yy}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \langle\delta_{yy}v,\delta_{xx}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}\right)\end{aligned}$$

Using summation by parts (16e), (16f) on the mixed terms in the above expression leads to

$$\begin{aligned}\kappa^2\langle\delta_{\Delta\boxplus}v,\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}_{N_x,N_y}} &= \kappa^2\nu\langle\delta_{\Delta\boxplus}v,\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}_{N_x,N_y}} + \kappa^2(1-\nu)\left(\langle\delta_{xx}v,\delta_{xx}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \langle\delta_{yy}v,\delta_{yy}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}\right)\\ &\quad + 2\kappa^2(1-\nu)\langle\delta_{x+y+}v,\delta_{x+y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \mathfrak{b}'\\ &= \delta_{t+}\mathfrak{v}' - \mathfrak{b}'\end{aligned}$$

where

$$\begin{aligned}\mathfrak{v}' &= \frac{\kappa^2\nu}{2}\langle\delta_{\Delta\boxplus}u,e_{t-}\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \frac{\kappa^2(1-\nu)}{2}\left(\langle\delta_{xx}u,e_{t-}\delta_{xx}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \langle\delta_{yy}u,e_{t-}\delta_{yy}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}\right)\\ &\quad + \kappa^2(1-\nu)\langle\delta_{x+y+}u,e_{t-}\delta_{x+y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}}\\ \mathfrak{b}' &= \kappa^2(1-\nu)\{\delta_{t\cdot}\delta_{y\cdot}u,\delta_{xx}u\}'_{(\mathbb{U}_{N_x},0)} + \kappa^2(1-\nu)\{\delta_{t\cdot}u,\delta_{xx}\delta_{y\cdot}u\}'_{(\mathbb{U}_{N_x},0)} + \kappa^2(1-\nu)\{\delta_{t\cdot}\delta_{x\cdot}u,\delta_{yy}u\}'_{\left(0,\mathbb{U}_{N_y}\right)}\\ &\quad + \kappa^2(1-\nu)\{\delta_{t\cdot}u,\delta_{x\cdot}\delta_{yy}u\}'_{\left(0,\mathbb{U}_{N_y}\right)} - \kappa^2(1-\nu)\{\delta_{t\cdot}\delta_{y\cdot}u,\delta_{xx}u\}'_{(\mathbb{U}_{N_x},N_y)} - \kappa^2(1-\nu)\{\delta_{t\cdot}u,\delta_{xx}\delta_{y\cdot}u\}'_{(\mathbb{U}_{N_x},N_y)}\\ &\quad - \kappa^2(1-\nu)\{\delta_{t\cdot}\delta_{x\cdot}u,\delta_{yy}u\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} - \kappa^2(1-\nu)\{\delta_{t\cdot}u,\delta_{x\cdot}\delta_{yy}u\}'_{\left(N_x,\mathbb{U}_{N_y}\right)} - 2\kappa^2(1-\nu)\delta_{t\cdot}u_{0,0}\delta_{x\cdot}\delta_{y\cdot}u_{0,0}\\ &\quad + 2\kappa^2(1-\nu)\delta_{t\cdot}u_{0,N_y}\delta_{x\cdot}\delta_{y\cdot}u_{0,N_y} + 2\kappa^2(1-\nu)\delta_{t\cdot}u_{N_x,0}\delta_{x\cdot}\delta_{y\cdot}u_{N_x,0} - 2\kappa^2(1-\nu)\delta_{t\cdot}u_{N_x,N_y}\delta_{x\cdot}\delta_{y\cdot}u_{N_x,N_y}\end{aligned}$$

and finally, from (32), to

$$\delta_{t+}\mathfrak{h} = \mathfrak{b} \tag{34}$$

where $\mathfrak{h} = \mathfrak{t} + \mathfrak{v}$, with $\mathfrak{v} = \mathfrak{v}_0 + \mathfrak{v}'$ of the form

$$\begin{aligned}\mathfrak{v} &= \frac{\gamma^2}{2}\langle\delta_{x+}u,e_{t-}\delta_{x+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{\gamma^2h_y}{4}\{\delta_{x+}u,e_{t-}\delta_{x+}u\}_{(\underline{\mathbb{U}}_{N_x},0)} + \frac{\gamma^2h_y}{4}\{\delta_{x+}u,e_{t-}\delta_{x+}u\}_{(\underline{\mathbb{U}}_{N_x},N_y)}\\ &\quad + \frac{\gamma^2}{2}\langle\delta_{y+}u,e_{t-}\delta_{y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{\gamma^2h_x}{4}\{\delta_{y+}u,e_{t-}\delta_{y+}u\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} + \frac{\gamma^2h_x}{4}\{\delta_{y+}u,e_{t-}\delta_{y+}u\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\\ &\quad + \frac{\kappa^2\nu}{2}\langle\delta_{\Delta\boxplus}u,e_{t-}\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \frac{\kappa^2(1-\nu)}{2}\left(\langle\delta_{xx}u,e_{t-}\delta_{xx}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \langle\delta_{yy}u,e_{t-}\delta_{yy}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}\right)\\ &\quad + \kappa^2(1-\nu)\langle\delta_{x+y+}u,e_{t-}\delta_{x+y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}}\\ &\quad - \frac{\sigma_1k}{2}\|\delta_{t-}\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} + \frac{\sigma_1h_yk}{4}\|\delta_{t-}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},0)} - \frac{\sigma_1h_yk}{4}\|\delta_{t-}\delta_{x+}u\|^2_{(\underline{\mathbb{U}}_{N_x},N_y)}\\ &\quad - \frac{\sigma_1k}{2}\|\delta_{t-}\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} + \frac{\sigma_1h_xk}{4}\|\delta_{t-}\delta_{y+}u\|^2_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} - \frac{\sigma_1h_xk}{4}\|\delta_{t-}\delta_{y+}u\|^2_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}\end{aligned}$$

and where $\mathfrak{b} = \mathfrak{b}_0 + \mathfrak{b}'$ is of the form

$$
\begin{aligned}
\mathfrak{b} = &-\gamma^2 \left\{\delta_{t.}u, \delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},0)} - \gamma^2 \left\{\delta_{t.}u, \delta_{x.}u\right\}'_{(0,\mathbb{U}_{N_y})} + \gamma^2 \left\{\delta_{t.}u, \delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},N_y)} + \gamma^2 \left\{\delta_{t.}u, \delta_{x.}u\right\}'_{(N_x,\mathbb{U}_{N_y})} \\
&+ \kappa^2 \left\{\delta_{t.}u, \delta_{y.}\left(\left(2-\nu\right)\delta_{xx} + \delta_{yy}\right)u\right\}'_{(\mathbb{U}_{N_x},0)} - \kappa^2 \left\{\delta_{t.}\delta_{y.}u, \left(\nu\delta_{xx} + \delta_{yy}\right)u\right\}'_{(\mathbb{U}_{N_x},0)} \\
&+ \kappa^2 \left\{\delta_{t.}u, \delta_{x.}\left(\delta_{xx} + \left(2-\nu\right)\delta_{yy}\right)u\right\}'_{(0,\mathbb{U}_{N_y})} - \kappa^2 \left\{\delta_{t.}\delta_{x.}u, \left(\delta_{xx} + \nu\delta_{yy}\right)u\right\}'_{(0,\mathbb{U}_{N_y})} \\
&- \kappa^2 \left\{\delta_{t.}u, \delta_{y.}\left(\left(2-\nu\right)\delta_{xx} + \delta_{yy}\right)u\right\}'_{(\mathbb{U}_{N_x},N_y)} + \kappa^2 \left\{\delta_{t.}\delta_{y.}u, \left(\nu\delta_{xx} + \delta_{yy}\right)u\right\}'_{(\mathbb{U}_{N_x},N_y)} \\
&- \kappa^2 \left\{\delta_{t.}u, \delta_{x.}\left(\delta_{xx} + \left(2-\nu\right)\delta_{yy}\right)u\right\}'_{(N_x,\mathbb{U}_{N_y})} + \kappa^2 \left\{\delta_{t.}\delta_{x.}u, \left(\delta_{xx} + \nu\delta_{yy}\right)u\right\}'_{(N_x,\mathbb{U}_{N_y})} \\
&- 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},0)} - 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{x.}u\right\}'_{(0,\mathbb{U}_{N_y})} + 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{y.}u\right\}'_{(\mathbb{U}_{N_x},N_y)} + 2\sigma_1 \left\{\delta_{t.}u, \delta_{t-}\delta_{x.}u\right\}'_{(N_x,\mathbb{U}_{N_y})} \\
&- 2\kappa^2\left(1-\nu\right)\delta_{t.}u_{0,0}\delta_{x.}\delta_{y.}u_{0,0} + 2\kappa^2\left(1-\nu\right)\delta_{t.}u_{0,N_y}\delta_{x.}\delta_{y.}u_{0,N_y} + 2\kappa^2\left(1-\nu\right)\delta_{t.}u_{N_x,0}\delta_{x.}\delta_{y.}u_{N_x,0} \\
&- 2\kappa^2\left(1-\nu\right)\delta_{t.}u_{N_x,N_y}\delta_{x.}\delta_{y.}u_{N_x,N_y}
\end{aligned}
$$

Clearly, the boundary term vanishes under any of the following choices of numerical boundary conditions at one of the four edges:

$$
\begin{aligned}
u_{0\le l\le N_x,0} = \delta_{x.}u_{0\le l\le N_x,0} &= \delta_{y.}u_{0\le l\le N_x,0} = 0 \\
u_{0,0\le m\le N_y} = \delta_{x.}u_{0,0\le m\le N_y} &= \delta_{y.}u_{0,0\le m\le N_y} = 0 \\
u_{0\le l\le N_x,N_y} = \delta_{x.}u_{0\le l\le N_x,N_y} &= \delta_{y.}u_{0\le l\le N_x,N_y} = 0 \\
u_{N_x,0\le m\le N_y} = \delta_{x.}u_{N_x,0\le m\le N_y} &= \delta_{y.}u_{N_x,0\le m\le N_y} = 0
\end{aligned}
\qquad \text{clamped} \tag{35a}
$$

$$
\begin{aligned}
u_{0\le l\le N_x,0} = \delta_{xx}u_{0\le l\le N_x,0} &= \delta_{yy}u_{0\le l\le N_x,0} = 0 \\
u_{0,0\le m\le N_y} = \delta_{xx}u_{0,0\le m\le N_y} &= \delta_{yy}u_{0,0\le m\le N_y} = 0 \\
u_{0\le l\le N_x,N_y} = \delta_{xx}u_{0\le l\le N_x,N_y} &= \delta_{yy}u_{0\le l\le N_x,N_y} = 0 \\
u_{N_x,0\le m\le N_y} = \delta_{xx}u_{N_x,0\le m\le N_y} &= \delta_{yy}u_{N_x,0\le m\le N_y} = 0
\end{aligned}
\qquad \text{simply supported} \tag{35b}
$$

$$
\begin{aligned}
\left(\nu\delta_{xx} + \delta_{yy}\right)u_{0\le l\le N_x,0} &= \left(\gamma^2\delta_{y.} - \kappa^2\delta_{y.}\left(\delta_{xx} + \left(2-\nu\right)\delta_{yy}\right) + 2\sigma_1\delta_{t-}\delta_{y.}\right)u_{0\le l\le N_x,0} = 0 \\
\left(\delta_{xx} + \nu\delta_{yy}\right)u_{0,0\le m\le N_y} &= \left(\gamma^2\delta_{x.} - \kappa^2\delta_{x.}\left(\delta_{xx} + \left(2-\nu\right)\delta_{yy}\right) + 2\sigma_1\delta_{t-}\delta_{x.}\right)u_{0,0\le m\le N_y} = 0 \\
\left(\nu\delta_{xx} + \delta_{yy}\right)u_{0\le l\le N_x,N_y} &= \left(\gamma^2\delta_{y.} - \kappa^2\delta_{y.}\left(\delta_{xx} + \left(2-\nu\right)\delta_{yy}\right) + 2\sigma_1\delta_{t-}\delta_{y.}\right)u_{0\le l\le N_x,N_y} = 0 \\
\left(\delta_{xx} + \nu\delta_{yy}\right)u_{N_x,0\le m\le N_y} &= \left(\gamma^2\delta_{x.} - \kappa^2\delta_{x.}\left(\delta_{xx} + \left(2-\nu\right)\delta_{yy}\right) + 2\sigma_1\delta_{t-}\delta_{x.}\right)u_{N_x,0\le m\le N_y} = 0
\end{aligned}
\qquad \text{free} \tag{35c}
$$

together with the corner conditions

$$
\delta_{x.}\delta_{y.}u_{0,0} = \delta_{x.}\delta_{y.}u_{0,N_y} = \delta_{x.}\delta_{y.}u_{N_x,0} = \delta_{x.}\delta_{y.}u_{N_x,N_y} = 0 \tag{36}
$$

To determine the conditions under which the numerical energy $\mathfrak{h}$ is positive definite, we start with showing that

$$
\begin{aligned}
&\left\langle \delta_{x+}u, e_{t-}\delta_{x+}u\right\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{h_y}{2}\left\{\delta_{x+}u, e_{t-}\delta_{x+}u\right\}_{(\underline{\mathbb{U}}_{N_x},0)} + \frac{h_y}{2}\left\{\delta_{x+}u, e_{t-}\delta_{x+}u\right\}_{(\underline{\mathbb{U}}_{N_x},N_y)} \\
&= \sum_{l=0}^{N_x-1}\sum_{m=0}^{N_y-1} h_xh_y\left(\delta_{x+}u_{l,m}\right)e_{t-}\delta_{x+}u_{l,m} - \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,0}\right)e_{t-}\delta_{x+}u_{l,0} + \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,N_y}\right)e_{t-}\delta_{x+}u_{l,N_y} \\
&= \sum_{l=0}^{N_x-1}\sum_{m=1}^{N_y-1} h_xh_y\left(\delta_{x+}u_{l,m}\right)e_{t-}\delta_{x+}u_{l,m} + \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,0}\right)e_{t-}\delta_{x+}u_{l,0} + \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{x+}u_{l,N_y}\right)e_{t-}\delta_{x+}u_{l,N_y} \\
&\overset{(4c)}{\ge} -\frac{k^2}{4}\sum_{l=0}^{N_x-1}\sum_{m=1}^{N_y-1} h_xh_y\left(\delta_{t-}\delta_{x+}u_{l,m}\right)^2 - \frac{h_xh_yk^2}{8}\sum_{l=0}^{N_x-1}\left(\delta_{t-}\delta_{x+}u_{l,0}\right)^2 - \frac{h_xh_yk^2}{8}\sum_{l=0}^{N_x-1}\left(\delta_{t-}\delta_{x+}u_{l,N_y}\right)^2 \\
&\overset{(18a)}{\ge} -\frac{k^2}{h_x^2}\|\delta_{t-}u\|^2_{\mathbb{U}^2_{N_x,N_y}}
\end{aligned}
$$

and likewise

$$\begin{aligned}\langle \delta_{y+}u, e_{t-}\delta_{y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}} - \frac{h_x}{2}\left\{\delta_{y+}u, e_{t-}\delta_{y+}u\right\}_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} + \frac{h_x}{2}\left\{\delta_{y+}u, e_{t-}\delta_{y+}u\right\}_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)}&\\ \overset{(4c)}{\geq} -\frac{k^2}{4}\sum_{l=1}^{N_x-1}\sum_{m=0}^{N_y-1} h_xh_y\left(\delta_{t-}\delta_{y+}u_{l,m}\right)^2 - \frac{h_xh_yk^2}{8}\sum_{m=0}^{N_y-1}\left(\delta_{t-}\delta_{y+}u_{0,m}\right)^2 - \frac{h_xh_yk^2}{8}\sum_{m=0}^{N_y-1}\left(\delta_{t-}\delta_{y+}u_{N_x,m}\right)^2&\\ \overset{(18b)}{\geq} -\frac{k^2}{h_y^2}\|\delta_{t-}u\|^2_{\mathbb{U}^2_{N_x,N_y}}&\end{aligned}$$

Along similar lines, we find

$$\begin{aligned}-\|\delta_{t-}\delta_{x+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} + \frac{h_y}{2}\|\delta_{t-}\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},0\right)} - \frac{h_y}{2}\|\delta_{t-}\delta_{x+}u\|^2_{\left(\underline{\mathbb{U}}_{N_x},N_y\right)}&\\ = -\sum_{l=0}^{N_x-1}\sum_{m=0}^{N_y-1} h_xh_y\left(\delta_{t-}\delta_{x+}u_{l,m}\right)^2 + \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{t-}\delta_{x+}u_{l,0}\right)^2 - \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{t-}\delta_{x+}u_{l,N_y}\right)^2&\\ = -\sum_{l=0}^{N_x-1}\sum_{m=1}^{N_y-1} h_xh_y\left(\delta_{t-}\delta_{x+}u_{l,m}\right)^2 - \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{t-}\delta_{x+}u_{l,0}\right)^2 - \frac{h_xh_y}{2}\sum_{l=0}^{N_x-1}\left(\delta_{t-}\delta_{x+}u_{l,N_y}\right)^2&\\ \overset{(18a)}{\geq} -\frac{4}{h_x^2}\|\delta_{t-}u\|^2_{\mathbb{U}^2_{N_x,N_y}}&\end{aligned}$$

and likewise

$$-\|\delta_{t-}\delta_{y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}} + \frac{h_x}{2}\|\delta_{t-}\delta_{y+}u\|^2_{\left(0,\underline{\mathbb{U}}_{N_y}\right)} - \frac{h_x}{2}\|\delta_{t-}\delta_{y+}u\|^2_{\left(N_x,\underline{\mathbb{U}}_{N_y}\right)} \overset{(18b)}{\geq} -\frac{4}{h_y^2}\|\delta_{t-}u\|^2_{\mathbb{U}^2_{N_x,N_y}}$$

For the remaining terms of $\mathfrak{v}$ we can write

$$\begin{aligned}\frac{\kappa^2\nu}{2}\langle \delta_{\Delta\boxplus}u, e_{t-}\delta_{\Delta\boxplus}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}&\\ + \frac{\kappa^2\left(1-\nu\right)}{2}\left(\langle \delta_{xx}u, e_{t-}\delta_{xx}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \langle \delta_{yy}u, e_{t-}\delta_{yy}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + 2\langle \delta_{x+y+}u, e_{t-}\delta_{x+y+}u\rangle_{\underline{\mathbb{U}}^2_{N_x,N_y}}\right)&\\ \overset{(4c)}{\geq} -\frac{\kappa^2k^2\nu}{8}\left(\|\delta_{t-}\delta_{\Delta\boxplus}u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2&\\ - \frac{\kappa^2k^2\left(1-\nu\right)}{8}\left[\left(\|\delta_{t-}\delta_{xx}u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 + \left(\|\delta_{t-}\delta_{yy}u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 + 2\|\delta_{t-}\delta_{x+y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}\right]&\\ = -\frac{\kappa^2k^2}{8}\left[\left(\|\delta_{t-}\delta_{xx}u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 + \left(\|\delta_{t-}\delta_{yy}u\|'_{\mathbb{U}^2_{N_x,N_y}}\right)^2 + 2\left(1-\nu\right)\|\delta_{t-}\delta_{x+y+}u\|^2_{\underline{\mathbb{U}}^2_{N_x,N_y}}\right]&\\ - \frac{\kappa^2k^2\nu}{8}\left(\langle \delta_{t-}\delta_{xx}u, \delta_{t-}\delta_{yy}u\rangle'_{\mathbb{U}^2_{N_x,N_y}} + \langle \delta_{t-}\delta_{yy}u, \delta_{t-}\delta_{xx}u\rangle'_{\mathbb{U}^2_{N_x,N_y}}\right)&\end{aligned}$$

References

[Bil09] Stefan Bilbao. *Numerical Sound Synthesis: Finite Difference Schemes and Simulation in Musical Acoustics*. John Wiley & Sons Ltd, 2009. ISBN: 9780470510469.